PH 101

Lecture 10

OUTLINE

- Lagrange equations of motion
- Exploring Lagrange equations of motion in *various systems*

Lagrange's Equation of Motion

According to the principle of least action a system takes that trajectory for which the action **S,**

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t$$

takes the extremum, generally the **minimum** value

$$\delta S = \delta \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t = 0,$$

Resulting finally in:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0}$$

When the system has more than one degree of freedom, the n different functions $q_i(t)$ must be varied independently in the principle of least action. We then evidently obtains equations of the form:

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0} \qquad (i = 1,2,\ldots,n) \qquad \textbf{(1)}$$

These are the required differential equations, called in mechanics Lagrange's equation

If the Lagrangian of a given mechanical system is known, the equations (1) give the relations between accelerations, velocities and coordinates, i.e. they are the equations of motion of the system.

Lagrange's Equation of Motion from Euler-Lagrange's equation

In Lecture 6 we learnt that:

If the function y(x) makes the integral functional

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x]dx$$

stationary, then y(x) must satisfy the Euler-Lagrange differential equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0$$

In the context of Lagrangian formalism, we can clearly see:

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x]dx \qquad \rightarrow S = \int_{t_1}^{t_2} L[q(t), \dot{q}(t), t]dt$$

So, here: $$f = L[q(t), \dot{q}(t), t]$$

Quite naturally,

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0 \qquad \Rightarrow \boldsymbol{\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = 0}$$

The Lagrangian function

For a conservative system, the Lagrangian of a system is defined as:

$$L(\boldsymbol{q}, \dot{\boldsymbol{q}}) = T(\boldsymbol{q}, \dot{\boldsymbol{q}}) - V(\boldsymbol{q})$$

T: Kinetic energy; V: Potential energy

Newton's Equation of Motion
Lagrange's Equation of Motion

Consider a particle of mass 'm' moving in 3-D in a potential $V(\vec{r})$. Using cartesian coordinates as generalized coordinates:

$$q_1 = x, \quad q_2 = y, \quad q_3 = z$$

We write:

$$T = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2 \right)$$

$$V = V(x, y, z)$$

$$T = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2 \right)$$

$$V = V(x, y, z)$$

So,

$$L = L(x, y, z\,;\ \dot{x}, \dot{y}, \dot{z})$$

$$= \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2 \right) - V(x, y, z)$$

The Lagrange's equations of motion for each generalized coordinates are:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{z}}\right) - \frac{\partial L}{\partial z} = 0$$

It is easy to obtain:

$$m\ddot{x} = -\frac{\partial V}{\partial x} = F_x \quad \rightarrow (i)$$

$$m\ddot{y} = -\frac{\partial V}{\partial y} = F_y \quad \rightarrow (ii)$$

$$m\ddot{z} = -\frac{\partial V}{\partial z} = F_z \quad \rightarrow (iii)$$

Obviously we can write the above equations in the compact form:

$$\vec{F} = m\ddot{\vec{r}} = m\vec{a}$$

The Simple Pendulum Problem

Consider the motion of the pendulum in Fig. 1

It is constrained to move in the xy plane along an arc of radius r. We could choose to describe the configuration of the pendulum by means of its position vector

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Clearly, such a choice is bad because it ignores the two conditions of constraint to which the pendulum must adhere, namely:

$$z = 0 \qquad r^2 - (x^2 + y^2) = 0$$

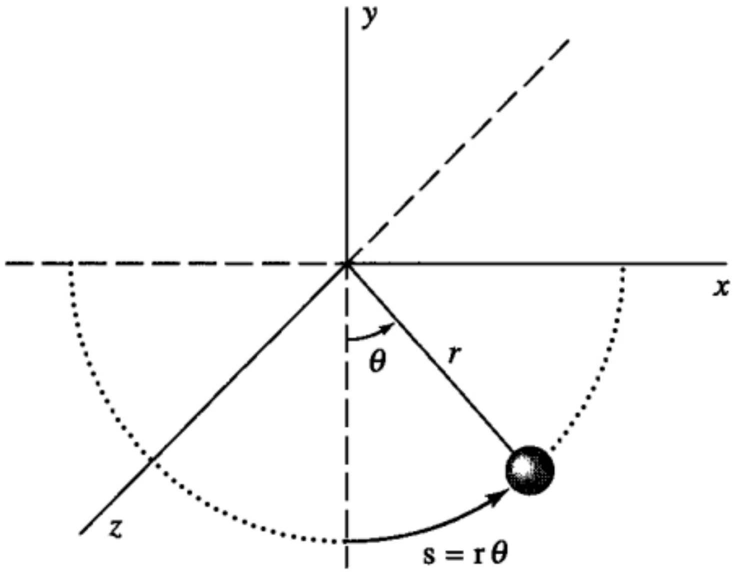


Fig. 1

Only one scalar coordinate is really needed to the position of the pendulum. At first sight either x or y might work, but to resolve a possible left—right ambiguity, we would probably choose x to define the position uniquely. We would then assume that the implied value of y is always negative. Given the x-value of the pendulum, y and z would then be determined completely by the conditions of constraint. **This choice would still prove awkward for describing the configuration of the pendulum.**

A better choice would be the arc length displacements s $(= r\theta)$ or, equivalently, the angular displacement θ of the pendulum away from the vertical. Either of these choices is better, because only a single number is needed to tell us the whereabouts of the pendulum. The important point here is that the pendulum really has only one degree of freedom; that is, it can move only one way and that way is along an arc of radius r. There exists only a single, independent coordinate necessary to depict its configuration uniquely.

When using the Lagrangian method, it is often useful to begin with rectangular or Cartesian coordinates and transform to the most obvious system with the simplest generalized coordinates. This will be our ground rule!

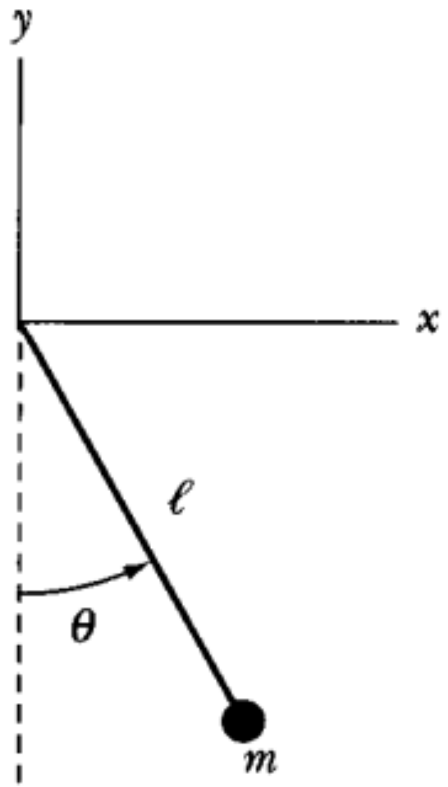


Fig. 2

In the case of simple pendulum of length l and bob of mass m, the kinetic and potential energies and the Lagrangian become:

$$T = \frac{1}{2} m\dot{x}^2 + \frac{1}{2} m\dot{y}^2$$

$$V = mgy$$

$$L = T - V = \frac{1}{2} m\dot{x}^2 + \frac{1}{2} m\dot{y}^2 - mgy$$

As already explained, the motion can be better expressed by using θ and $\dot{\theta}$

Transforming x and y into coordinates θ

$$x = \ell \sin \theta$$
$$y = -\ell \cos \theta$$

We now find for $\dot{x}$ and $\dot{y}$

$$\dot{x} = \ell\dot{\theta} \cos \theta$$

$$\dot{y} = \ell\dot{\theta} \sin \theta$$

$$L = \frac{m}{2}(\ell^2\dot{\theta}^2 \cos^2 \theta + \ell^2\dot{\theta}^2 \sin^2 \theta) + mg\ell \cos \theta = \frac{m}{2}\ell^2\dot{\theta}^2 + mg\ell \cos \theta$$

$\boxed{\text{The only generalized coordinate in the case of the pendulum is the angle } \theta}$

The equations of motion are as follows:

$$\frac{\partial L}{\partial \theta} = -mg\ell \sin\theta$$

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = m\ell^2\ddot{\theta}$$

Thus,

$$\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0$$

The Double Pendulum Problem

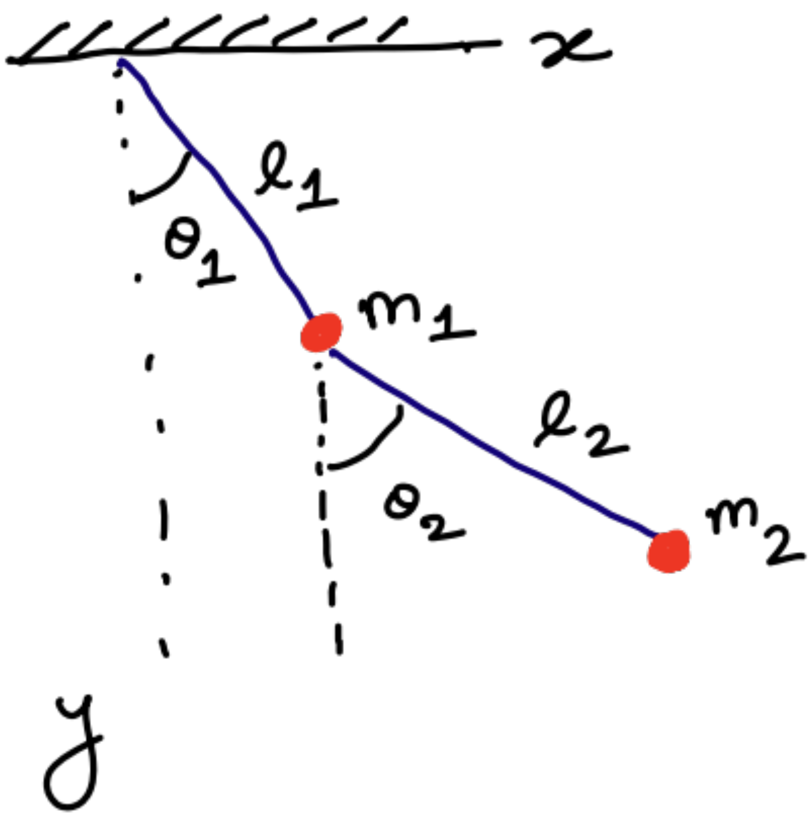
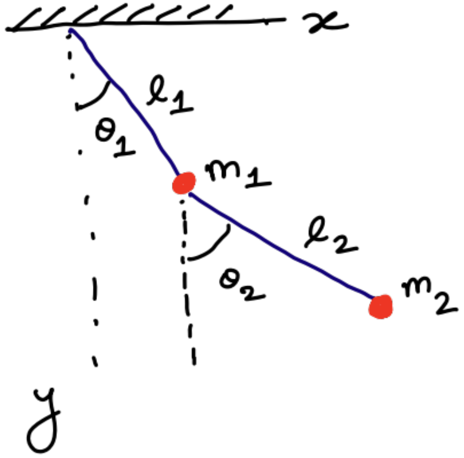


Fig. 3

Generalized coordinates in this problem are θ_1 and θ_2.

For the particle m_1:

$$x_1 = l_1 \sin\theta_1$$
$$y_1 = l_1 \cos\theta_1$$



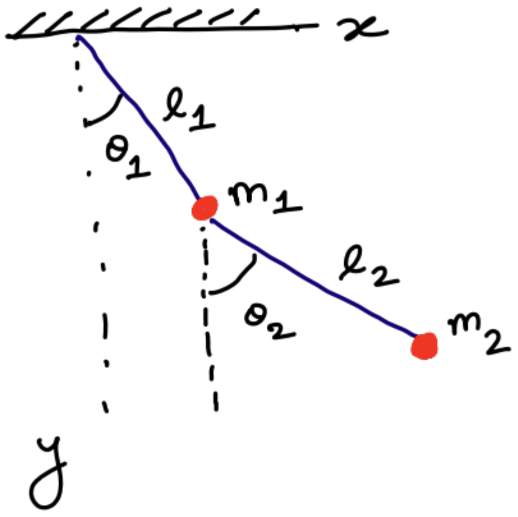
$$T_1 = \frac{1}{2} m_1 \left(\dot{x}_1^2 + \dot{y}_1^2\right) \quad ; \quad V_1 = - m_1 g y_1$$

$$\Rightarrow \boxed{T_1 = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2} \qquad \Rightarrow \boxed{V_1 = - m_1 g l_1 \cos \theta_1}$$

For particle m_2:

$$x_2 = x_1 + l_2 \sin \theta_2 = l_1 \sin\theta_1 + l_2 \sin\theta_2$$

$$y_2 = y_1 + l_2 \cos\theta_2 = l_1 \cos\theta_1 + l_2 \sin\theta_2$$



$$T_2 = \frac{1}{2} m_2 \left(\dot{x}_2^2 + \dot{y}_2^2 \right)$$

$$= \frac{1}{2} m_2 \left[l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \cos(\theta_1 - \theta_2) \dot{\theta}_1 \dot{\theta}_2 \right]$$

$$V_2 = -m_2 g y_2$$

$$= - m_2 g \left(l_1 \cos\theta_1 + l_2 \cos\theta_2 \right)$$

Thus,

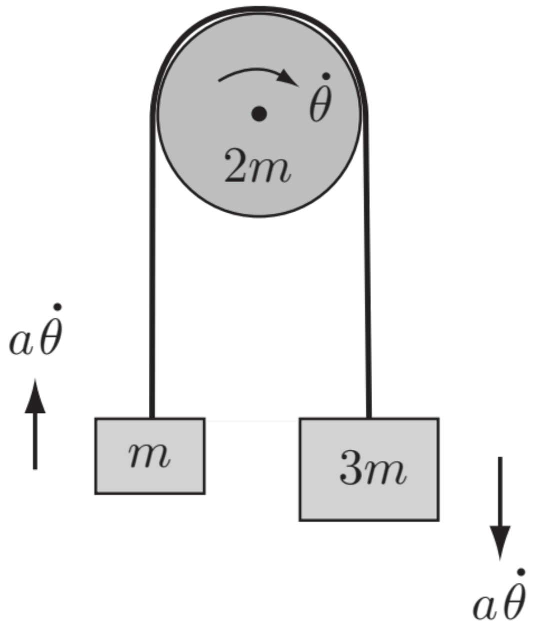
$$L = \frac{1}{2} (m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2$$

$$+ m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2)$$

$$+ (m_1 + m_2) l_1 \cos\theta_1 + m_2 g l_2 \cos\theta_2$$

The Atwood Machine Problem

A uniform circular pulley of mass $2m$ can rotate freely about its axis of symmetry which is fixed in a horizontal position. Two masses m, $3m$ are connected by a light inextensible string which passes over the pulley without slipping. The whole system undergoes planar motion with the masses moving vertically. Take the rotation angle of the pulley as generalised coordinate and obtain Lagrange's equation for the motion. Deduce the upward acceleration of the mass m.



Solution:

Let θ be the rotation angle of the pulley measured from some reference configuration. Then the velocity diagram is shown in Fig. 4 The kinetic energy of the system is

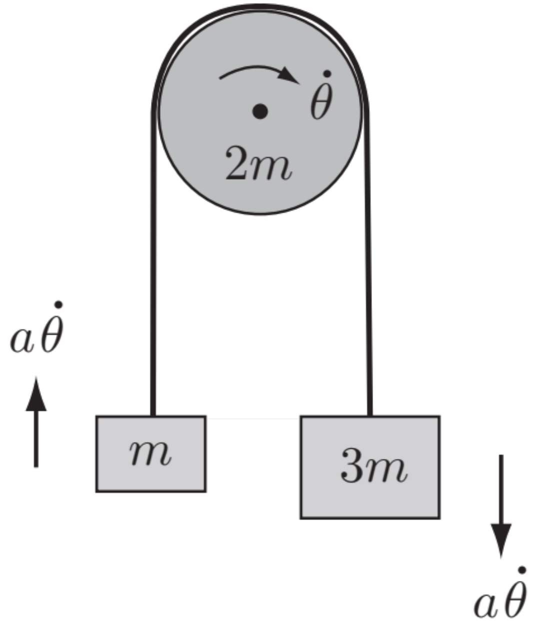


Fig. 4

$$T = \tfrac{1}{2}m\left(a\dot{\theta}\right)^2 + \tfrac{1}{2}(3m)\left(a\dot{\theta}\right)^2 + \tfrac{1}{2}\left(\tfrac{1}{2}(2m)a^2\right)\dot{\theta}^2$$
$$= \tfrac{5}{2}ma^2\dot{\theta}^2$$

and the potential energy relative to the reference configuration is

$$V = mg\,(a\theta) + (3m)g\,(-a\theta)$$
$$= -2mga\theta.$$

Lagrange's equation for the system is therefore

$$\frac{d}{dt}\left(5ma^2\dot{\theta}\right) - 0 = 2mga,$$

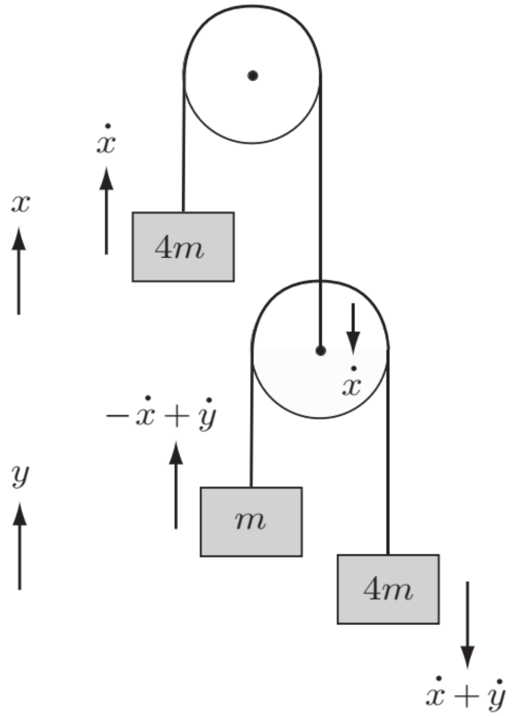
that is,

$$a\ddot{\theta} = \tfrac{2}{5}g.$$

The **upward acceleration** of the mass m is therefore $\frac{2}{5}g$. ■

Double Atwood Machine Problem

A light pulley can rotate freely about its axis of symmetry which is fixed in a horizontal position. A light inextensible string passes over the pulley. At one end the string carries a mass $4m$, while the other end supports a second light pulley. A second string passes over this pulley and carries masses m and $4m$ at its ends. The whole system undergoes planar motion with the masses moving vertically. Find Lagrange's equations and deduce the acceleration of each of the masses.



Solution:

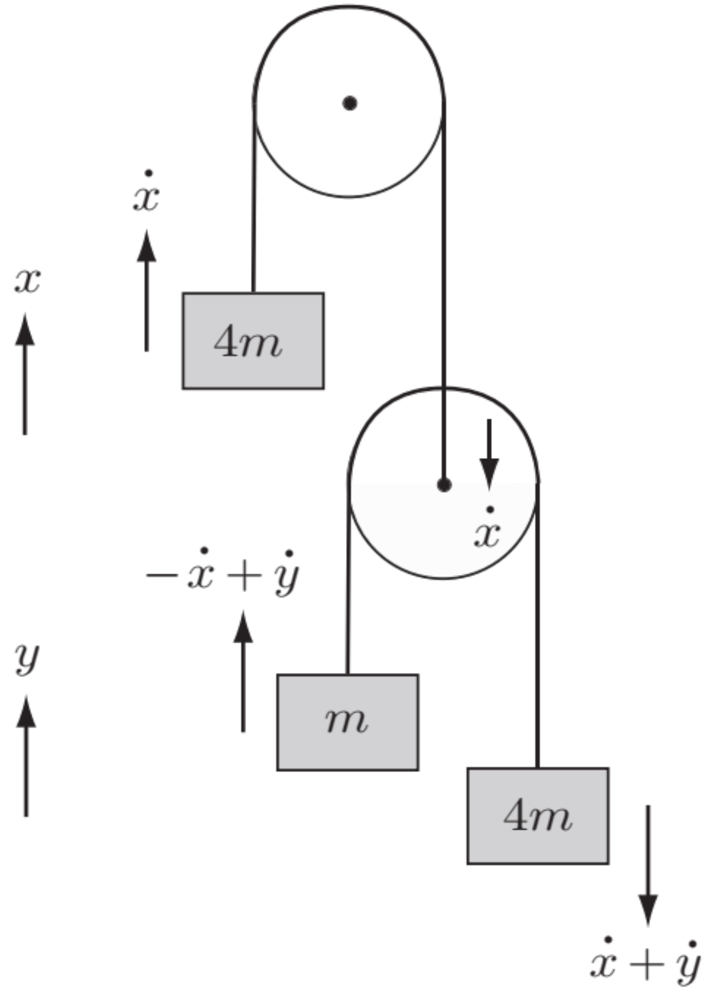


Fig. 5

Let x be the upward displacement of the first mass $4m$, and let y be the upward displacement of the mass m measured relative to the centre of the lower pulley. Then the velocity diagram is shown in Fig. 5. The **kinetic energy** of the system is

$$\begin{aligned} T &= \tfrac{1}{2}(4m)\dot{x}^2 + \tfrac{1}{2}m\,(-\dot{x}+\dot{y})^2 + \tfrac{1}{2}(4m)\,(\dot{x}+\dot{y})^2 \\ &= \tfrac{1}{2}m\left(9\dot{x}^2 + 6\dot{x}\,\dot{y} + 5\dot{y}^2\right) \end{aligned}$$

and the **potential energy** is

$$\begin{aligned} V &= (4m)gx + mg(-x+y) - 4mg(x+y) \\ &= -mgx - 3mgy. \end{aligned}$$

Lagrange equations for the system are therefore

$$\frac{d}{dt}(9\dot{x} + 3\dot{y}) = g,$$
$$\frac{d}{dt}(3\dot{x} + 5\dot{y}) = 3g,$$

that is,

$$9\ddot{x} + 3\ddot{y} = g,$$
$$3\ddot{x} + 5\ddot{y} = 3g.$$

These simultaneous linear equations have the solution

$$\ddot{x} = -\tfrac{1}{9}g, \qquad \ddot{y} = \tfrac{2}{3}g.$$

The **accelerations** of the three masses are therefore $\frac{1}{9}g$ downwards, $\frac{7}{9}g$ upwards, and $\frac{5}{9}g$ downwards. ■